Marinade

Inside Marinade's staking stack

An introductory tour of staking infrastructure on *Solana*

Ondra Chaloupka

What we are going to cook through

What is staking

Why a blockchain pays you to lock money up

Liquid staking

The original Solana LST

Native staking

Keep custody. No program ever holds your SOL.

Bonds and instant unstake

Collateral, and skipping the cooldown

Who talks to you

Ondra Chaloupka @_chalda

Backend developer at Marinade

Before that, Java engineer at Red Hat Red Hat

Came for distributed systems, stayed for Solana

Contributor to Realms, and author of its SPL Governance deep dive

Who is Marinade

Born in a hackathon, 2021

Two projects merged, and shipped the first liquid staking token on Solana.

A DAO with a public forum

MNDE holders lock their tokens to vote, and proposals are argued in the open.

A stake automation platform

Marinade's own words. Chasing rewards and Solana's decentralisation at once.

Five years on one chain, bootstrapped by grants rather than venture capital.

What is staking

Backing the network

Locked SOL says its security and uptime matter, and the network pays for that.

Validators do the work

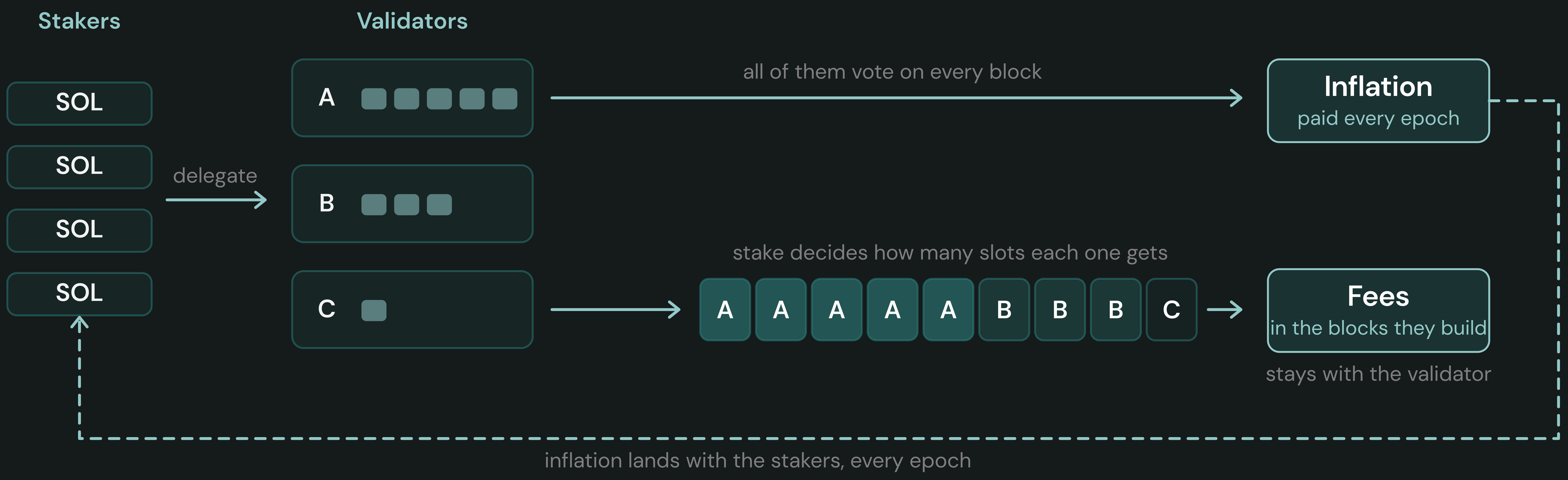
They run the hardware and vote on blocks, carrying the stake delegated to them.

Two kinds of reward

Voting earns inflation. Building blocks earns fees.

Solana's own wording: delegating your tokens does not give the validator ownership or control over them.

Stake decides who builds



Base fee is 5,000 lamports per signature, half burned and half to the validator. Priority fees go to the validator in full.

Liquid staking

You all know what liquid staking is

You hand over SOL

The protocol takes it. Only you can ask for your share back.

An on-chain program holds it

Marinade delegates it, and keeps delegating it well.

You hold mSOL

Liquidity you can use across DeFi.

First liquid staking protocol on Solana. 2021, out of two hackathon teams.

Stake · Bond · Auction · Settle

You staked.

Now what?

Somebody has to choose the validators

Watch

Every validator on Solana, every epoch.

Judge

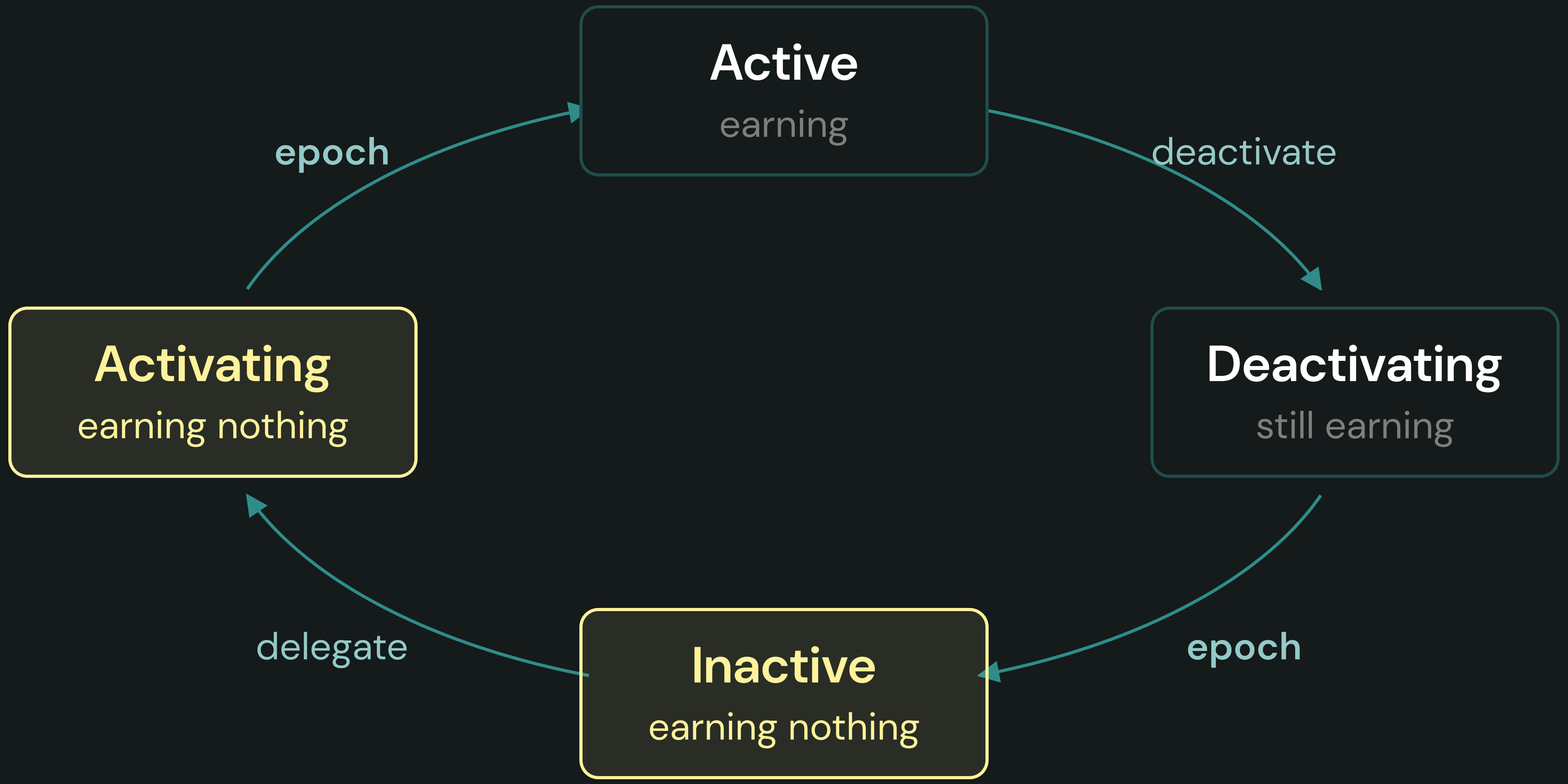
Who pays best, who stays up, who keeps the network spread out.

Move

Stake follows the answer.

A validator worth staking last epoch may not be worth it in this one.

A stake account cannot move sideways



There is no arrow across the middle. Solana never enabled redelegation.

Bonds. Validators back their word with their own SOL.

Stake — Bond — Auction — Settle

Validators bid for your stake

1

Bid

A validator offers a share of its rewards.

2

Allocate

Highest first, until the stake runs out.

3

Clear

The last winner in sets the price.

Everyone who wins is paid that same clearing rate, not their own bid. Inflation and MEV come from Solana. This is the third thing you earn.

From promise to payment

Until SIMD-0123 is live, Solana has no native way to pass priority fees to stakers.

1. **Measure**
   End of epoch, read what happened.
2. **Calculate**
   What each validator owes.
3. **Settle**
   Written on chain, out of the bonds.
4. **Claim**
   Bids and covered losses reach the stakers.

Not everyone wants a program holding their SOL

Native staking

No contract risk

The SOL never leaves your possession.

No token

Nothing to hold or explain to an auditor.

Just the delegation

Marinade picks the validators, nothing more.

Launched July 2023. Rewards land straight in your account each epoch, so it compounds without anyone doing anything.

Solana splits the keys

```
pub enum StakeStateV2 {
    Uninitialized, Initialized(Meta), Stake(Meta, Stake, StakeFlags), RewardsPool,
}
```

```
Meta            rent_exempt_reserve, authorized, lockup
  Authorized    staker, withdrawer
  Lockup        unix_timestamp, epoch, custodian
Stake           delegation, credits_observed
  Delegation    voter_pubkey, stake, activation_epoch, deactivation_epoch
```

```
pub struct Authorized {
    pub staker: Pubkey,     // may delegate
    pub withdrawer: Pubkey, // may take the money
}
```

One account. Custody is in Meta, the delegation is in Stake, and Marinade only ever holds one field of the first.

Nobody at Marinade holds a keyring like this.

The staking authority is a program address.

Stake

Exit

Three ways to run it

Max Yield

The default. Your stake follows the winners of the auction.

Select

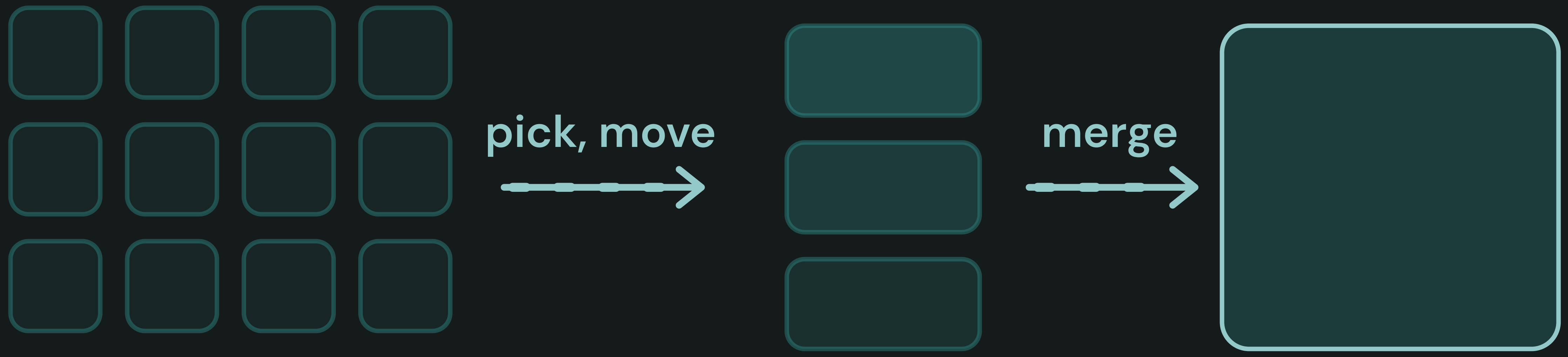
A curated, identity-verified set. Built for institutions.

Recipes

Your rewards are swapped, epoch by epoch, into a token you pick.

Recipes pays out in $USDG $ZBTC $MNDE $BONK $FWOG $NOBODY $TRENCHER $USDC

Getting out is the hard part



Spread over many validators

Deactivated in batches

One withdrawal

Moving an account to the exit authority is what marks it as leaving, so the authority is the state. A hundred accounts does not fit in one transaction.

Instant unstake

Somebody buys your stake account

You hand over the stake account → They hand over SOL

One transaction. Both sides settle, or neither does. No liquid token, no cooldown, and it works on any stake account, even ones Marinade never touched.

Somebody has to want it

You get
SOL now, a little under what the account holds.

They get
Your stake account, and the wait that comes with it.

The gap
The difference between the two is the price of not waiting.

No unstaking fee from Marinade, and you see the price before you sign.

Marinade knows how this works

We watch every validator, every epoch.

We turn a promise into collateral you can read.

We get you out again.

The machinery is temporary. That is the point.

Stake · Bond · Auction · Settle · Exit

Marinade

Stake it till you *make* it

github.com/marinade-finance
docs.marinade.finance

Appendix, if it comes up

MEV arrives as a bundle

An ordered group

Transactions run in the order given, inside one slot, all of them or none.

A tip, not a priority fee

Bundles compete on tips to the validator, paid on top of the normal fee.

Split on chain

A merkle root per validator per epoch, a capped commission, and stakers claim the rest.

Marinade's auction already prices it in: expected MEV is part of what a validator can share back.